Algorithms and Responsibility

Felix Bönisch*

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Abstract

A long line of experimental work suggests that decision-makers can—and do—use delegation to shift responsibility for unwelcome outcomes onto an intermediary. We ask whether the same logic carries over to delegation to algorithms. In a preregistered online experiment ($N = 322$ UK Prolific participants, 161 in each role), a decision-maker makes a payoff-relevant prediction for a third party with the option to delegate to an algorithm calibrated, at the individual level, to perform as well as she does. We manipulate, between subjects, whether the third party can punish her for the decision. Two findings emerge. First, contrary to the preregistered prediction of the responsibility-avoidance literature, the *prospect* of punishment *reduces* rather than increases the likelihood of delegating to the algorithm (a 17 percentage-point gap, two-sided $p = 0.033$). Second, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions; the point estimate runs in the predicted direction but the test is underpowered to rule out moderate insulation effects. We discuss several candidate mechanisms for the headline reversal and argue that, among those we can evaluate, a process-ownership account—in which the moral salience introduced by the punishment possibility raises the value of being the proximate cause of a good outcome—is the most consistent with the data, while flagging that the supporting evidence is exploratory and the design cannot rule out experimenter demand. Within the bounds of the setting we study, our results offer a behavioural complement to regulatory frameworks that hold human users accountable for algorithm-mediated decisions: such frameworks may discipline algorithmic adoption rather than merely allocate post-hoc responsibility.

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*WZB Berlin Social Science Center. Email: `felix.boenisch@wzb.eu`. Acknowledgements to be added at submission.

1 Introduction

Algorithms are increasingly involved in consequential decisions: they screen job candidates, set credit scores, recommend medical treatments, evaluate parole applicants, and—more mundanely but at vast scale—route the cars driving toward this very page. As the locus of decision-making shifts toward algorithmic systems, a parallel question has moved into public, legal, and academic debate: when an algorithm-mediated decision turns out badly, who bears the blame? A natural intuition, which finds support in a long line of experimental work on responsibility avoidance through delegation, is that decision-makers can use algorithms as a means to shift blame onto a non-human intermediary. To return to a familiar example: if you and a friend arrive late because she relied on a navigation app rather than her own sense of direction, the algorithm seems a convenient scapegoat—and the next time she is unsure of the route, the prospect of avoiding your blame may itself motivate her use of the app. This paper provides experimental evidence on whether either of these intuitions is borne out.

We address two questions. *First*: can decision-makers actually shift responsibility away from themselves by delegating to an algorithm—that is, are they punished less for a delegated decision than for a self-made decision with the same outcome? *Second*: does the prospect of being held responsible motivate the decision to delegate in the first place? The two questions correspond to, respectively, the *viability* and the *motivational salience* of algorithmic responsibility avoidance. They have been studied in different settings and rarely in combination, and previous designs have struggled to identify them cleanly. We do so here in a preregistered online experiment.

Our experimental design pins down both questions. Player A first completes ten rounds of an incentivised visual prediction task (estimating a person's weight from a facial image and stated height). She then makes an eleventh prediction whose outcome determines the payoff of a second participant, Player B: a high payoff if the prediction falls within ten pounds of the true weight, a low payoff otherwise. Player A may either make this prediction herself or delegate it to an algorithm. The algorithm is calibrated to mimic Player A's individual performance from the first ten rounds—that is, on the eleventh prediction it produces the high outcome with the same probability as Player A herself. Player A is told this. We manipulate the punishment possibility between subjects: in the *Punishment* condition, Player B may reduce Player A's payoff conditional on the delegation decision and the realised outcome; in the *No-Punishment* condition, no punishment is possible. Player B's punishment behaviour is elicited via the strategy method across all four (delegation, outcome) cells, providing a within-subject test of whether delegation insulates Player A from punishment.

Two design features distinguish our setting from the closest precedent, Feier et al. (2022). First, the algorithm is calibrated at the *individual* level rather than the session or group level. Existing designs typically expose subjects to a distribution of algorithmic performance and let them infer their own ability relative to that distribution; the delegation decision then carries strong signal value about the principal's self-perception, and any differential punishment can reflect overconfidence rather than responsibility shifting per se. By neutralising relative-performance signalling at the individual level, we isolate the responsibility-shifting motive. Second, our *No-Punishment* condition turns off the punishment possibility entirely, allowing us to identify the effect of anticipated punishment on the delegation decision itself—a test that is not feasible in designs where punishment is always possible. Both features sharpen the test of responsibility shifting in algorithmic delegation that the literature has thus far been able to deliver.

Two findings emerge, and in their joint structure they push against the natural reading of the responsibility-avoidance literature. *First, the prospect of punishment reduces, rather than increases, the likelihood of delegation:* 42.5% of Player As delegate when punishment is possible, compared to 59.3% when it is not (a 17 percentage-point gap; two-sided Pearson Chi-squared $p = 0.033$, Fisher exact $p = 0.041$; the preregistered one-sided test in the H1 direction is non-significant by construction, since the result reverses the predicted direction—we discuss this in detail in Section 4). This rejection of the standard prediction in favour of the opposite direction is the paper's main empirical contribution, and we view it as a behavioural counterpart to the regulatory intuition that holding humans accountable for algorithm-mediated decisions disciplines their use of those algorithms. *Second, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions in our setting:* Player B's punishment of Player A is statistically indistinguishable across the two, and Player A correctly anticipates this absence of differential punishment in her elicited beliefs. The point estimate of any insulation effect runs in the predicted direction but the test is underpowered against moderate alternatives ($n \approx 67$ Player Bs after attention-check exclusion); the broader claim that responsibility shifting through algorithmic delegation is altogether absent is one we cannot make with this data.

What mechanism reconciles the two findings? Among the candidates we can evaluate with our design, a *process-ownership* account is the most consistent with the data—though we treat it as a candidate hypothesis rather than an established mechanism, since the supporting evidence is exploratory and a fourth candidate (experimenter demand) cannot be ruled out by construction. A simple anticipation story—in which Player A delegates less when punishment is on the table because she expects harsher punishment for delegated decisions—is inconsistent with the elicited beliefs and with Player B's actual choices. A simple effort story—in which Player A makes the prediction herself in order to outperform the algorithm and earn the high payoff—is inconsistent with the data, in which non-delegators in the *No-Punishment* condition outperform non-delegators in the *Punishment*. What remains is the milder hypothesis that, when the moral salience of the decision is heightened by the punishment possibility, Player A wants to be the proximate cause of any good outcome for Player B, irrespective of any quantifiable belief about how she will be judged. Consistent with this interpretation, the negative correlation between Player A's task performance and her propensity to delegate is observed in the *Punishment* condition only: those most likely to deliver the high payoff want to do so personally when the moral stakes are made salient. Section 4 weighs the four candidate mechanisms in detail and is explicit about what the data can and cannot rule out.

These findings contribute to three strands of literature. First, we add to the experimental literature on responsibility avoidance through delegation (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffel et al., 2016), by extending the canonical setting to delegation to an algorithmic intermediary in a setting with genuine outcome uncertainty. Second, we contribute to the rapidly growing literature on algorithm aversion and adoption (Dietvorst et al., 2015; Logg et al., 2019; Burton et al., 2020; Mahmud et al., 2022; Chugunova and Sele, 2022) by isolating an under-studied determinant of algorithm use: the institutional context in which the human user is held accountable. Third, and most directly, we build on Feier et al. (2022) and Kirchkamp and Strobel (2019) by providing a clean experimental test of two long-standing claims in the responsibility-and-algorithms literature, with a design that controls relative-performance signalling at the individual level and that includes a condition in which punishment is ruled out by construction. The contrast with Feier et al. (2022) is consequential: that study finds that decision-makers are held *less* responsible for algorithmically delegated decisions, while we find that the punishment possibility *reduces* delegation rather than increasing it. We argue in Section 2 that the two findings are reconcilable: the relative-performance signalling that drives much of Feier et al.'s differential punishment is closed off by construction in our design, isolating a different and milder margin (the motivational salience of accountability for the principal). Beyond the experimental literature, our findings speak to the regulatory debate over the assignment of responsibility for algorithmic decisions (European Union, 2024): within the bounds of the setting we study, holding human users accountable for algorithm-mediated outcomes appears to be a behavioural lever for direct human engagement with the decision, rather than merely a tool for ex-post moral attribution. We are explicit in Section 5 about why the present evidence—a single online experiment on a stylised prediction task with one UK Prolific sample—supports this claim only in a bounded sense, and what would be required to extend it to the high-stakes algorithmic domains (medical, judicial, hiring, financial) that motivate the regulatory debate.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature in detail. Section 3 describes the experimental design and states our two preregistered hypotheses. Section 4 presents the empirical findings, including the mechanism analysis. Section 5 discusses the implications and limitations of our results.

2 Related Literature

This paper bridges three strands of literature: (i) responsibility avoidance through delegation, an established line of experimental work in which the intermediary is another human; (ii) algorithm aversion and adoption, which examines when and why people are willing to use algorithmic decision aids; and (iii) responsibility attributions to decisions involving algorithms, an emerging literature at the intersection of the first two. We review each in turn and conclude by clarifying the gap our experiment addresses.

Responsibility avoidance through delegation

The political-science and public-choice literatures have long recognised that delegation can serve as a strategy for shifting blame onto an intermediary (Weaver, 1986; Fiorina, 1986). Experimental economics has established the same point in incentivised settings, almost always using modified dictator games.

Bartling and Fischbacher (2012) provide the canonical experimental demonstration. In their design, a dictator may either choose between a fair and an unfair allocation of an endowment or instead delegate this choice to an intermediary whose monetary incentives are aligned with the dictator's. The recipient is adversely affected if the unfair allocation is chosen and may punish the dictator and the intermediary. Punishment is effectively shifted from the dictator to the intermediary, and the prospect of avoiding punishment constitutes a strong motive for the delegation decision itself. Hamman et al. (2010) document a complementary pattern: dictators are more generous when they allocate directly than when they select an intermediary from a set of candidates. Qualitative responsibility measures align with the punishment patterns—dictators feel less responsible when they delegate, and recipients place less blame on principals than on intermediaries.

Coffman (2011) replicates the reduction in punishment under delegation but reinterprets the mechanism. He finds that punishment decreases not because of any inherent shifting of blame to the intermediary, but because the dictator no longer "directly interacts" with the affected party. This holds even when the intermediary is powerless. Oexl and Grossman (2013) extend the design to include various restrictions on the intermediary's choice set and confirm that punishment can be successfully avoided through delegation even to a powerless intermediary, in a setting where punishment is costly, the intermediary's incentives are aligned with the dictator's, and the intermediary can also be punished. Together, the two studies suggest that the lack of direct interaction between dictator and recipient is a key channel through which delegation reduces punishment.

Hill (2015) provides a vignette-based corroboration in a legislative-delegation context. Subjects directly evaluate blame in unincentivised hypothetical scenarios; the resulting attributions track the patterns observed in incentivised experiments, suggesting that punishment captures more qualitative blame attributions reliably. Steffel et al. (2016) push the literature in a complementary direction, showing that people are more willing to delegate decisions made on behalf of others than decisions made for themselves, especially when those decisions might have negative outcomes; the asymmetry is driven both by anticipated responsibility and by anticipated blame, and people care more about avoiding blame for bad outcomes than about claiming credit for good ones. Erat (2013) provides a related demonstration that people are willing to pay to delegate the act of lying, consistent with delegation reducing the psychological cost of behaviour that one expects to be judged negatively.

These studies share a common conclusion: in settings with self-interested choice between a fair and an unfair allocation, delegation reduces punishment of the principal, and the prospect of punishment in turn motivates delegation. Their design space, however, has two important features that constrain external validity to algorithmic settings. First, the intermediary is human, and the principal can typically be punished for choosing a particular kind of intermediary. Second, the underlying choice is between a known fair and a known unfair allocation, with no inherent uncertainty in the outcome. Whether responsibility avoidance is just as effective when the intermediary is an algorithm and the underlying decision involves genuine uncertainty is the question we take up. We turn next to what is known about delegation when the intermediary is algorithmic.

Delegation to algorithms

A large literature studies when people use algorithmic decision aids. Dietvorst et al. (2015) introduced the term "algorithm aversion" for the finding that people prefer their own (or other humans') judgement over algorithmic predictions, even after observing that algorithms outperform human decision-makers. Subsequent work has documented many moderators of this preference. Algorithm aversion is stronger when the algorithm cannot be observed to learn from its mistakes (Berger et al., 2021), when the algorithm's reasoning is opaque (Litterscheidt and Streich, 2020; Sharan and Romano, 2020), when the algorithm is presented as more complex (Stein et al., 2020), and—most importantly for our purposes—when the underlying decision has moral content or affects third parties (Gogoll and Uhl, 2018; Bigman and Gray, 2018; Jauernig et al., 2022). Algorithm aversion is also stronger when the task is perceived as more uncertain (Dietvorst and Bharti, 2020), when delegation removes any ability to override the algorithm's choice (Dietvorst et al., 2018), and when the algorithm is used in tasks perceived as subjective (Castelo et al., 2019; Bogert et al., 2021). Logg et al. (2019), by contrast, document conditions under which subjects show *appreciation* of algorithmic advice over human advice; the picture that emerges is therefore one of substantial heterogeneity in algorithm preferences across tasks, contexts, and elicitation methods. Burton et al. (2020), Mahmud et al. (2022), and Chugunova and Sele (2022) provide systematic reviews of this literature.

Within this body of work, a smaller subset focuses on settings in which the decision being delegated has consequences for someone other than the decision-maker, sometimes called the *moral domain* (Gogoll and Uhl, 2018). Gogoll and Uhl (2018) find that subjects in a laboratory experiment preferred to delegate a simple calculation task to another participant rather than to an algorithm even when the task was a straightforward arithmetic exercise where algorithmic advantage should have been salient; subjects had observed the algorithm fail in 80% of cases, however, which complicates inference about pure preferences. Goldbach et al. (2019) report similar aversion when the decision affects both the decision-maker and a third party, and Niszczota and Kaszás (2020) document the same in a financial decision-making context. The consensus across this subset is that aversion to algorithms is amplified when third-party welfare is at stake. The mechanism, however, is unsettled: *lack of trust in the algorithm*, *loss of perceived* *control*, and *anticipated reputational or moral costs* have all been proposed, and existing designs typically cannot separate them.

Responsibility perceptions of decisions involving algorithms

A small but growing literature examines how decisions involving algorithms are evaluated by third parties. Most early work is hypothetical and unincentivised: subjects read scenarios involving human-algorithm pairs and rate responsibility on Likert scales. Kurtzberg and Naquin (2004) find that observers attribute less responsibility to a company when an accident involves technology. Pezzo and Pezzo (2006) find similarly that observers attribute less fault to a doctor who follows a (bad) recommendation from an automated decision aid. Liu and Du (2021) replicate this effect in a contemporary setting. Shank et al. (2019) consider moral violations produced by human-algorithm pairs and find that humans who monitor algorithms are blamed less than humans acting alone or humans receiving algorithm recommendations, suggesting that responsibility attributions are sensitive to the precise division of authority between human and machine. Bigman and Gray (2018) document a robust "algorithmic outrage asymmetry": moral violations by algorithms produce less outrage than identical violations by humans. Tobia et al. (2021) extend the discussion to liability law and argue that physicians may follow recommendations from automated systems precisely because doing so shields them from legal blame. The literature thus far suggests that responsibility attributions to humans involved with algorithms differ systematically from attributions to humans acting alone, but the direction depends on the exact configuration.

A smaller incentivised experimental literature has begun to test these ideas in laboratory settings with monetary stakes. Kirchkamp and Strobel (2019) compare a modified dictator game in which the dictator either acts alone or shares responsibility with a computer; they find no significant differences in self-reported responsibility between human-computer and human-human teams, although a (statistically insignificant) tendency towards more selfish allocations under shared responsibility with humans. The most closely related study is Feier et al. (2022), who study delegation to an algorithm in an incentivised laboratory experiment. In their design, subjects first complete ten logic tasks, are randomly assigned to either a "human" or "machine" treatment, and then either rely on their own performance or delegate to (respectively) another human or an algorithm to determine a third party's payoff. Third parties may then reward or punish. The headline finding is that decision-makers are held *less* responsible for decisions delegated to an algorithm than for decisions delegated to another human, conditional on outcome.

The Feier et al. (2022) design, which is the immediate precedent for ours, has three features that complicate the inference about pure responsibility shifting. First, the algorithm in their experiment is calibrated at the session level rather than at the individual level: each subject is shown the distribution of algorithmic performance across the relevant pool, and her delegation decision then reflects beliefs about her own ability *relative* to that distribution. The delegation choice therefore carries strong signal value about Player A's self-perception, which contaminates downstream punishment as a measure of responsibility shifting per se. Feier et al. (2022) indeed report that the delegation decision is well explained by perceived own performance relative to the algorithm. Second, the same subject acts as both delegator and evaluator, introducing additional motives—self-justification, hypocrisy aversion, projection—into the punishment decision. Third, their design does not include a treatment in which punishment is ruled out by construction, so the question of whether the punishment possibility itself motivates delegation cannot be addressed in their data.

Where this paper fits

Our experiment is the first to our knowledge to (i) calibrate the algorithm to mimic each individual subject's performance, eliminating the relative-performance confound; (ii) separate the role of delegator and evaluator across distinct subject pools; and (iii) include a between-subject manipulation that turns the punishment possibility on or off, providing a clean test of whether anticipated punishment is a motive in the delegation decision itself. The first two features sharpen the test of the responsibility-shifting hypothesis (H2) by removing alternative interpretations of differential punishment. The third moves beyond the existing literature, which has documented punishment differentials *conditional on* the punishment possibility but has not directly identified whether the threat of punishment causally moves the delegation decision.

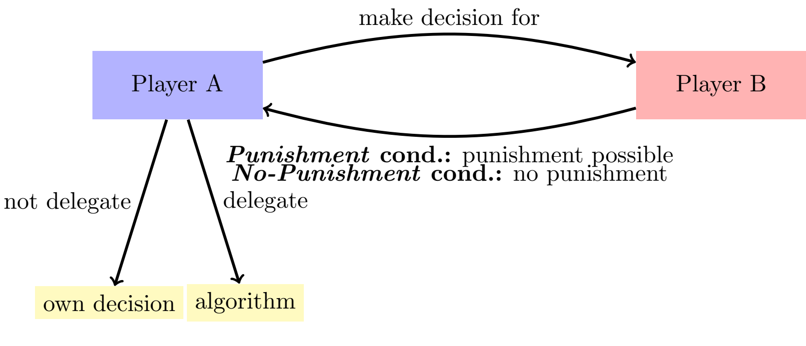
Reconciling our findings with Feier et al. (2022). Two of our results sit in apparent tension with the closest precedent. First, Feier et al. (2022) document that decision-makers are punished *less* for algorithmically delegated decisions; we find no significant punishment differential. Second, the natural prediction from Feier et al. (2022)'s own findings is that the prospect of punishment should make algorithmic delegation *more* attractive; we find the reverse. We do not view these contrasts as contradictions, but as evidence that the design features which differ between the two studies are doing real work. The differential punishment in Feier et al. (2022) is most plausibly read as third parties extracting a signal from the delegation decision about the principal's self-perceived ability relative to the algorithm—a signal that is maximally informative when (as in their design) the algorithm is calibrated at the session level and when the principal's relative ability is uncertain. Our individual-level calibration, made common knowledge, closes off this signal-extraction channel by construction: a Player B in our setting has no basis to read delegation as overconfidence, laziness, or ability avoidance. The disappearance of the punishment differential in our data is therefore consistent with, rather than refutes, the signal-extraction reading of Feier et al. (2022)'s result. By the same logic, the responsibility-shifting motive that Feier et al. (2022)'s setup leaves on the table is muted in ours, isolating a different and milder margin: the moral salience of accountability for the principal herself, independent of any third-party inference. The reversal we observe in the delegation decision speaks to that margin—and we are explicit that the question of which design feature is causally responsible for the sign reversal cannot be resolved without an experiment that varies relative-performance calibration directly, which we do not do here.

We see scope, then, for the two studies to be parts of a single picture: Feier et al. (2022) identifies the signal-extraction channel through which delegation can shift third-party blame; we identify a process-ownership channel through which the prospect of accountability can pull the principal toward direct engagement with the decision. The two channels are not mutually exclusive, and a unified design that varies both relative-performance calibration and the punishment possibility would be the natural next step. We discuss the implications for future work in Section 5.

3 Experimental Design

We conducted a preregistered online experiment via the British platform Prolific between February and June 2023, recruiting 322 participants from the United Kingdom. The experiment employs a between-subject manipulation of whether costly punishment is possible, combined with a within-subject elicitation of conditional punishment behavior. Within each condition, participants assumed one of two roles: Player A, who decides whether to make a payoff-relevant prediction for Player B directly or to delegate this decision to an algorithm calibrated to perform as well as Player A herself; and Player B, who—in the *Punishment* condition—can subsequently punish Player A based on both the delegation decision and the resulting outcome. In the *No-Punishment* condition, punishment is not available. We label the two between-subject conditions *Punishment* and *No-Punishment* to keep the relationship between the experimental manipulation and the condition name transparent throughout the paper. Figure 1 summarises the basic structure.

Figure 1: Conditions overview



Sample

We targeted a balanced design of 80 participants per role per condition, for a planned sample of 320 subjects, with the cell size set to detect the effects anticipated from the responsibility-avoidance literature at conventional power; the underlying ex-ante power calculation is reported in the preregistration document and reproduced in Appendix B. Due to attrition and subsequent re-matching, we ultimately recruited 322 participants (161 Player A's and 161 Player B's); a CONSORT-style flow diagram broken down by condition and role appears in Appendix C. The experiment was programmed in oTree (Chen et al., 2016) and restricted to participants completing the study on a computer rather than a mobile device, with Prolific's standard quality screens (minimum 95% prior approval rate, no warnings on file, fluent-English filter, located in the United Kingdom). Player A spent on average 12 minutes completing the experiment and earned an average bonus of £2.49 on top of the Prolific show-up fee; Player B spent on average 8 minutes and earned an average bonus of £2.30 on top of the show-up fee.[1]

Procedure

Player A

After providing informed consent, Player A completed ten incentivised rounds of a visual prediction task. In each round, the participant predicted a person's weight from a facial image and the person's height. Predictions could be entered via slider or typed directly in kilograms (kg), pounds (lbs), or stones and pounds (st, lbs). Participants did not receive feedback between rounds. At the end of the experiment, one round was randomly selected for payment: participants earned £4 if their prediction was within 10 lbs of the true weight, and £2 otherwise. The first ten rounds served three purposes: to give Player A a stake from which potential punishment could later be deducted; to familiarise her with the task; and to assess her individual performance for calibrating the algorithm. Participants were informed that all individuals depicted in the images had explicitly consented to the use of their photographs for research purposes.

We chose a visual prediction task for several reasons. Prior research suggests that individuals' willingness to use algorithms depends sensitively on the algorithm's perceived relative competence (Dietvorst et al., 2015; Logg et al., 2019). As described below, our design neutralises concerns about relative competence by calibrating the algorithm to match each individual subject's performance. The visual nature of the task reinforces the plausibility that an algorithm can err and lends credibility to our claim of equal performance, in contrast to purely mathematical tasks where participants may be skeptical of algorithms failing (Gogoll and Uhl, 2018). Predicting body weight from facial images has been used in prior experiments (Logg et al., 2019), and while in many tasks algorithms benefit from training on large datasets, our task arguably levels the playing field: humans have developed extensive real-life experience

[1] Bonus computed deterministically from cleaned data per the experimental rules: random-round task payoff (expectation over the uniform draw), belief-about-own-score payoff (£0.30 times the mass placed on the realized score), belief-about-distribution payoff (expected value over the uniform random draw of which score is checked), and—for Player A in the *Punishment* condition—expected punishment from the matched Player B's strategy-method choice in the realised cell, with £0.10 per cell of strategy-method punishment cost for Player B. The bonus excludes the risk-elicitation MPL payoff (raw-data column not retained in the cleaned files); including it would add roughly £0.30–£0.50 per subject.

mapping facial features to body weight. We deliberately chose a prediction task involving genuine uncertainty rather than a setting with an obvious self-serving motive (as in classic dictator-game variations). This choice reflects the nature of many real-world applications of algorithms—financial forecasting, medical diagnosis, fraud detection—in which outcomes are inherently uncertain and performance is evaluated ex post.

After the ten rounds, Player A was randomly paired with another participant, Player B. Player A was informed that her final, eleventh prediction would not affect her own payoff but would almost entirely determine the payoff of Player B, who had not completed the prediction task himself. Specifically, if the prediction fell within 10 lbs of the true weight, Player B would receive a high payoff (£5); otherwise, Player B would receive a low payoff (£1). Player A then learned that she could either make this prediction herself or delegate it to an algorithm. Crucially, Player A was informed that the algorithm had been calibrated to mimic her own performance from the first ten rounds and would therefore on expectation produce the high payoff for Player B with the same frequency as Player A herself.

Concretely, the algorithm draws a Bernoulli outcome with success probability equal to Player A's empirical accuracy rate over the first ten rounds. A subject who scored 3/10 in the initial rounds thus delegates to an algorithm with a 30% probability of producing the high payoff for Player B. This individual-level Bernoulli calibration is the central design innovation of our experiment relative to the closest precedent (Feier et al., 2022), and it does substantive work in the identification argument: it makes the equal-performance claim literally true on average for every subject, and it eliminates any signal value that the delegation decision might otherwise carry about Player A's beliefs concerning her own ability relative to the algorithm. A Player B observing the delegation decision thus has no basis on which to read it as overconfident, lazy, or selfish—which closes off the most plausible third-party signal-extraction channel through which delegation might otherwise insulate the principal from punishment.

By neutralising relative-performance concerns and adopting a neutral framing that avoids describing the algorithm's internal workings, we isolate the responsibility-shifting motive from the most prominent confound in the existing literature. Had the algorithm instead been calibrated to reflect aggregate performance across a group of participants—as is the case in the most closely related study, Feier et al. (2022)—delegation choices would be confounded by participants' beliefs about their own ability relative to the group. Feier et al. (2022) indeed find that the delegation decision in their setting is largely driven by perceived own performance relative to the algorithm.

In the *Punishment* condition, Player A was further informed that Player B could reduce her payoff by up to £2, with the punishment depending on both the delegation decision and the prediction outcome. In the *No-Punishment* condition, Player B had no punishment option. The two delegation options were presented in random order. If Player A chose not to delegate, she was asked to make the final prediction and to state her confidence in having earned the high payoff for Player B. This step was skipped if the algorithm made the prediction.

We then asked Player A to report her beliefs about Player B's punishment in each of the four scenarios resulting from the combination of delegation decision (yes/no) and prediction outcome (high/low payoff for Player B). Beliefs were elicited on the same scale as actual punishment (£0–£2 in £0.10 increments) and the four scenarios were presented in block-wise random order. We chose not to incentivise belief elicitation for three reasons. First, only two of the four scenarios remain payoff-relevant after the delegation decision, making the others purely hypothetical. Second, an incentivised mechanism could induce hedging behaviour. Third, an incentive-compatible scoring rule across four scenarios would have substantially increased task complexity. Because we are nonetheless interested in comparing punishment beliefs to actual punishment, we included an attention check on the belief-elicitation screen.[2] In the *No-Punishment* condition, no punishment was possible; we elicited beliefs about hypothetical punishment to maintain consistency across treatments.

Subsequently, Player A was asked to assess her own performance in the first ten rounds by assigning a probability mass (summing to 100%) to each possible score from 0 to 10. This elicitation was incentivised: Player A earned an additional £0.30 with the probability she had assigned to her actual score.[3] Player A was also asked to estimate the distribution of scores in the population by assigning shares to each possible score. At the end of the experiment, one score was randomly drawn and Player A earned £0.30 if her estimate for that score was within five percentage points of the actual share. Finally, we elicited risk preferences using an incentivised multiple price list in the style of Holt and Laury (2002) and administered a short questionnaire on socio-demographic characteristics and technology affinity, before presenting Player A with her preliminary payoff. Figure 6 summarises the experimental timeline for Player A.

Player B

After providing informed consent, Player B was informed that he had been randomly matched with another participant, Player A, and was introduced to the task Player A had completed. Player B learned that his own payoff would be determined by the outcome of Player A's eleventh prediction: £5 if the prediction fell within 10 lbs of the true weight and £1 otherwise. Player A's payoff, in turn, depended on her performance in the initial ten rounds. Player B was further informed that Player A had the option to delegate the eleventh prediction to an algorithm calibrated to perform equally well as Player A in the earlier rounds, and that Player A was aware of this performance equivalence.

In the *Punishment* condition, Player B could choose to reduce Player A's payoff by up to £2, at a personal cost of £0.10 per chosen punishment level.[4] We elicited punishment decisions for

[2] Subjects who failed the attention check are excluded from belief-related analyses, as preregistered.

[3] This mechanism is not incentive-compatible for risk-neutral participants. In practice only one subject assigned the full probability mass to a single score, suggesting that hedging considerations were modest.

[4] The cost is fixed per scenario rather than proportional to the punishment level, so that strictly positive punishment is always costly relative to no punishment but the cost does not increase with the severity of the

all four combinations of delegation (yes/no) and outcome (high/low payoff for Player B) using the strategy method, with only the decision corresponding to the actually realised combination being implemented. Because the punishment decision is central to our research question, we included an attention check on the punishment-elicitation screen.[5] In the *No-Punishment* condition, where punishment was not possible, we elicited hypothetical punishment decisions on the same scale, to maintain comparability across treatments. Subsequently, we collected Player B's perceptions of task difficulty, his risk preferences, and a short questionnaire, before presenting his final payoff. Figure 7 summarises the experimental timeline for Player B.

After all decisions of Player B had been recorded, Player A's final payoff was calculated and all participants were paid on the day of participation.

Hypotheses

Our experimental design tests two preregistered hypotheses, motivated by the literature on responsibility avoidance through delegation (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffel et al., 2016) and recent work on responsibility attributions to algorithmic decision-makers (Feier et al., 2022; Kirchkamp and Strobel, 2019; Gogoll and Uhl, 2018).

H1 (Delegation): The introduction of potential punishment increases the likelihood that Player A delegates the decision to the algorithm.

H2 (Punishment): Conditional on a low-payoff outcome for Player B, Player A is punished less when she delegated the decision to the algorithm than when she made it herself.

H1 is grounded in robust empirical evidence that responsibility avoidance shapes delegation decisions involving human intermediaries. Bartling and Fischbacher (2012) additionally demonstrate that participants are more likely to delegate morally consequential decisions even to a die roll—an early form of delegation to a non-intentional agent—suggesting that responsibility avoidance persists when the agent lacks intentionality. Feier et al. (2022) further find no significant difference in the propensity to delegate to a human versus an algorithmic agent, suggesting that the nature of the agent may matter less than the opportunity to shift responsibility.

H2 mirrors H1 on the punishment side. Bartling and Fischbacher (2012) find that delegation to a randomisation device still allows decision-makers to avoid some punishment, although less so than delegation to another human. Feier et al. (2022) show that decision-makers are rewarded less when a positive outcome is directly attributable to their own action, compared to when it results from an algorithm to which they delegated. Their study differs from ours

sanction.

[5]The attention check asked the subject to confirm a numerical aspect of the payoff structure that had been displayed on the previous screen. Subjects failing this check are excluded from punishment-related analyses, as preregistered.

in three respects, however: a different task; an emphasis on reward differentials rather than punishment; and—most consequentially—an experimental design that does not control for relative performance at the individual level. As a result, in their setting the delegation decision carries a strong signal about Player A's beliefs concerning her own ability relative to the algorithm. A failure to delegate may then be punished as overconfidence rather than as direct responsibility-taking, contaminating the inference about responsibility-shifting per se.

Identification

Our design pins down two distinct quantities. The within-subject comparison between the four (delegation, outcome) cells of Player B's strategy-method punishment isolates the conditional-on-outcome effect of delegation on punishment, holding fixed the realised consequences for Player B (H2). The between-subject comparison of Player A's delegation rate across *Punishment* and *No-Punishment* isolates the effect of the punishment possibility on the propensity to delegate, holding fixed every other feature of the decision environment (H1).

A natural concern is that, although the prediction task affects Player B's payoff, the incentives of Player A and Player B are nominally aligned: Player A has no direct financial interest in Player B receiving the low payoff, and a successful prediction strictly weakens punishment incentives.[6] A standard intention-based punishment motive should therefore have less bite here than in dictator-game settings where the decision-maker can choose to be selfish. We discuss the implications of this design choice for the interpretation of our results in Section 4.

4 Results

This section presents the main empirical findings. Section 4 describes the sample, attention-check pass rates, and condition balance. Section 4 establishes our headline finding: the introduction of potential punishment *reduces* the likelihood of delegating to the algorithm, the opposite of what the responsibility-avoidance literature predicts. Section 4 shows that, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions; we are explicit about what the underlying test can and cannot rule out. Section 4 examines whether beliefs about punishment are consistent with these patterns. Section 4 weighs four candidate mechanisms for the unexpected direction of Result 1, ruling two out and treating the remaining two—process ownership and experimenter demand—as candidate hypotheses we cannot fully separate with this data. A concise summary of all preregistered tests, and of every deviation from the preregistration, appears in Appendix B.

[6]We consider Player A's incentives to be aligned with Player B's because, in the *Punishment* condition, a better outcome for Player B reduces expected punishment for Player A.

Sample, attention, and balance

We collected data via the British online platform Prolific between February and June 2023, recruiting all participants from the United Kingdom and restricting eligibility to subjects completing the experiment on a desktop or laptop computer. Beyond the desktop restriction, eligibility followed Prolific's standard quality screens: minimum prior approval rate of 95%, no warnings on file, and a fluent-English filter. A total of 322 subjects participated (161 Player A and 161 Player B). Player A spent on average 12 minutes on the experiment and earned an average bonus of £2.49 on top of the Prolific show-up fee; Player B spent approximately 8 minutes and earned an average bonus of £2.30. Average performance in the first ten prediction rounds was 2.93 correct rounds out of ten, statistically indistinguishable across conditions (2.76 in *Punishment* vs 3.09 in *No-Punishment*; two-sided t-test $p > 0.10$).

Random assignment to condition produced statistically indistinguishable Player A subsamples across socio-demographic characteristics. Appendix Table 2 reports balance on age, gender, self-reported socio-economic status, university education, technology affinity, and leadership experience for Player A; Appendix Table 3 reports the parallel balance check for Player B.[7] A CONSORT-style flow diagram showing recruitment, attrition, attention-check exclusion, and the analysis sample, broken down by condition and role, appears in Appendix C; in particular, attrition rates do not differ meaningfully across the *Punishment* and *No-Punishment* conditions.

We preregistered exclusion of subjects who failed the attention check on the screen at which their main outcome variable was elicited. For Player A, this is the belief-elicitation screen; for Player B, the punishment-elicitation screen. Of 161 Player Bs, 135 pass the attention check (83.9%); the corresponding pass rate for Player A is 131/161 (81.4%). All main results are robust to including subjects who fail the attention check; we report these robustness checks in Appendix A.3.

Result 1: Punishment reduces delegation

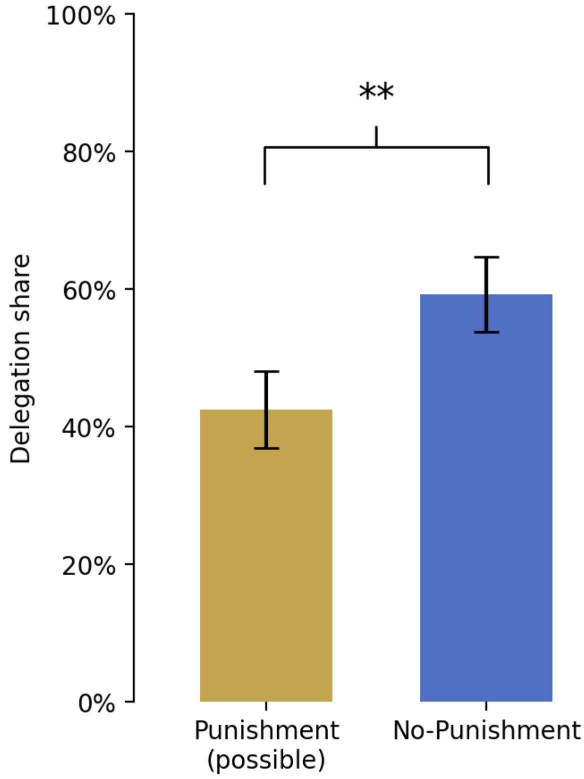
We begin with Player A's delegation decision. In the *Punishment* condition, where Player B can punish, 42.5% of Player As delegated the decision to the algorithm. In the *No-Punishment* condition, where punishment is impossible, 59.3% delegated. Figure 2 illustrates the difference. The introduction of potential punishment thus *reduces* the likelihood of delegation by approximately 17 percentage points; the difference is statistically significant at conventional levels under a two-sided test (Pearson Chi-squared, $p = 0.033$; Fisher's exact, $p = 0.041$).[8]

[7] Player B's punishment behaviour is identified within-subject across the four scenarios of the strategy method, so balance across conditions is not strictly required for the punishment analysis. We nonetheless report balance for completeness and to permit the reader to verify that the assignment-induced subsamples are comparable in observables.

[8] The preregistration specified a one-sided test in the H1 direction (more delegation under punishment). Because the realised effect reverses that direction, the preregistered one-sided test is non-significant by construction. Following standard practice for findings that reverse the predicted direction, we report two-sided tests throughout the body of the paper and treat all subsequent mechanism analyses as exploratory rather

Result 1. *The possibility of punishment significantly reduces the likelihood that Player A delegates the decision to the algorithm.*

Figure 2: Delegation shares by condition



Note: Bars show the share of Player As who delegated the eleventh prediction to the algorithm in each between-subject condition. The two-sided difference is significant at the 5% level under both a Pearson Chi-squared test (p = 0.033) and a Fisher's exact test (p = 0.041); the preregistered one-sided test in the H1 direction is non-significant by construction (the realised effect reverses the predicted sign).

This result not only fails to support *Hypothesis 1*; it points in the opposite direction. The responsibility-avoidance literature, on which H1 rests, would predict the introduction of punishment to make delegation *more* attractive as a tool for shifting responsibility. We instead observe a sizeable reduction in delegation when punishment is on the table. We discuss candidate mechanisms for this reversal in Section 4; for the remainder of this subsection we establish that the effect is robust and document its empirical structure.

Robustness. Table 1 reports estimates from a Logit model of the delegation decision. Column (1) reports the unconditional condition effect; Column (2) adds Player A's task performance and a vector of socio-demographic and technology-affinity controls. The condition effect is robust: the coefficient on the *No-Punishment* indicator is 0.68 (p = 0.034, two-sided) in the unconditional specification and 0.76 (p = 0.023, two-sided) with controls. The full set of estimated coefficients is reported in Appendix Table 4.

than confirmatory. The full set of preregistered tests, the deviations from each, and the rationale appears in

Table 1: Determinants of the delegation decision

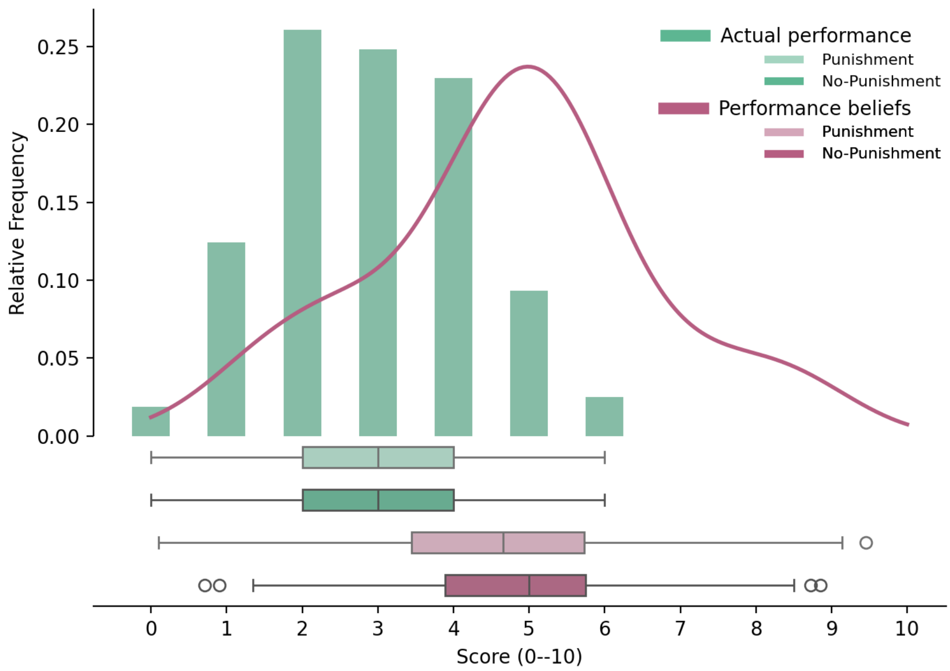
	Dependent variable: *Delegation*		
	Full sample		*Punishment only*
	(1)	(2)	(3)
No-Punishment indicator	0.677	0.757	
	(0.320)	(0.334)	
Punishment beliefs: bad outcome (no del. − del.)			0.937
			(0.751)
Punishment beliefs: good outcome (no del. − del.)			1.579
			(0.741)
Task performance		−0.192	−0.609
		(0.128)	(0.236)
Controls	no	yes	yes
N	161	159	60
Pseudo R^2	0.020	0.049	0.246

Note: Coefficients from a Logit regression of the delegation decision on the condition indicator, task performance, and socio-demographic controls. Robust (HC1) standard errors in parentheses. Two-sided p-values for the headline coefficients: *No-Punishment indicator* Col (1) $p = 0.034$; Col (2) $p = 0.023$; *Task performance* Col (3) $p = 0.010$; *Punishment beliefs: good outcome* Col (3) $p = 0.033$. Controls in Columns (2) and (3) include age, gender, self-reported socio-economic status, university education, technology affinity, and leadership experience; full results in Appendix Table 4. Two subjects are dropped in Column (2) due to missing socio-demographic data. Column (3) restricts to the *Punishment* condition and includes within-subject differences in expected punishment between non-delegated and delegated decisions for each outcome; subjects who failed the belief-screen attention check are excluded, as preregistered. Following AEA editorial policy, significance stars are not reported in the table; exact p-values are stated in this note.

Heterogeneity by performance. Within the *Punishment* condition, Player As who performed better on the first ten rounds were less likely to delegate. The Pearson correlation between the score in the first ten rounds and the probability of delegation is -0.19 (two-sided $p = 0.092$); the corresponding regression coefficient in Column (3) of Table 1 is -0.61 (two-sided $p = 0.010$). The strengthening of the relationship in the regression reflects, in part, the attention-check restriction (Column (3) is identified off $n = 60$ Baseline-attention-pass subjects, while the raw correlation uses all 80 Baseline subjects). In the *No-Punishment* condition, no comparable relationship between performance and delegation is observed. The pattern suggests that the condition effect is concentrated among Player As who, conditional on punishment being on the table, prefer to take a personal stake in producing a good outcome for Player B—a hint that we develop in Section 4. A binned scatter that visualises this heterogeneity is in preparation and will be added once the analysis pipeline is in place; the underlying numbers are reported in Table 1.

Calibration of the task. Delegation shares around 50% in both conditions are reassuring evidence that we chose a task in which subjects on average credibly perceive their own performance and that of the algorithm as comparable. The mean score on the first ten rounds is 2.93, statistically indistinguishable across conditions (Figure 3). Cross-condition differences in the propensity to delegate therefore do not stem from *ex-ante* differences in performance, nor—as established below—from differential effort exerted on the eleventh prediction.

Figure 3: Performance in the first ten rounds, by treatment



Note: Distribution of correct predictions (within 10 lbs of the true weight) over the first ten task rounds, by treatment. Average performance is 3.05 correct out of ten and is statistically indistinguishable across treatments.

Appendix B.

Result 2: We cannot reject equal punishment of delegated and self-made decisions

We now turn to Player B's punishment behaviour in the *Punishment* condition. Throughout the analysis we exclude subjects who failed the attention check on the punishment-elicitation screen, as preregistered; all results are robust to including these subjects (Appendix A.3, full robustness panel pending the analysis pipeline).

Punishment in all four (delegation, outcome) cells of the strategy method is significantly greater than zero. Player Bs punish significantly more for low-payoff outcomes than for high-payoff outcomes, indicating that punishment carries informational content rather than being noise; 41.8% of Player Bs nonetheless do not punish in any of the four scenarios. Conditional on the outcome, however, Player A is not significantly differently punished depending on the delegation decision. When Player B receives the high payoff, average punishment is £0.311 for delegated decisions and £0.310 for self-made decisions—a difference of one-tenth of a penny. When Player B receives the low payoff, average punishment is £0.494 when Player A delegated versus £0.571 when she made the decision herself; this difference points in the H2 direction but is far from significant (paired t-test $p = 0.153$; Wilcoxon signed-rank $p = 0.533$). Figure 4 summarises these patterns. The same picture obtains when measuring punishment frequencies rather than amounts (Appendix Figure 8): Player Bs neither punish delegated decisions more harshly, nor do they punish them more frequently.

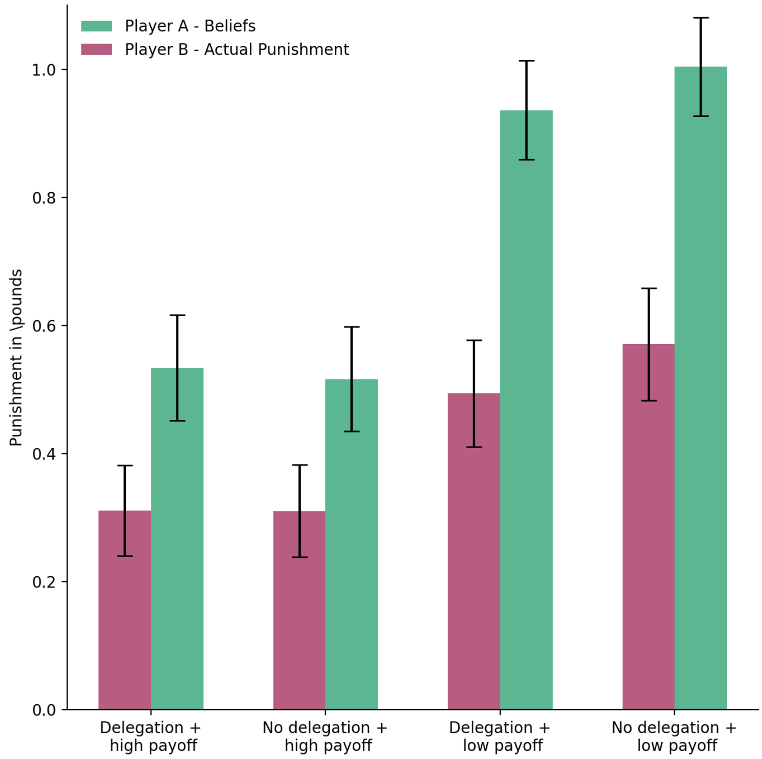
We frame this result with caution. The within-subject test on the bad-outcome cells is identified off approximately 67 Player Bs (the 135 attention-check passers, restricted to the *Punishment* condition). At this sample size, the test is well-powered against large insulation effects but underpowered against modest ones; the H2-direction point estimate of £0.077 has a wide confidence interval that we cannot interpret as ruling out moderate insulation. We therefore state Result 2 as a failure to reject equal punishment, not as evidence that punishment is invariant to delegation, and we report the corresponding minimum-detectable-effect calibration in Appendix A.3 once the analysis pipeline is in place.

Result 2. *Conditional on the outcome for Player B, we cannot reject the hypothesis that Player A is punished equally for delegated and self-made decisions. The point estimate of any insulation effect runs in the H2 direction but is small relative to the test's confidence interval.*

What predicts punishment? A regression of chosen punishment on a delegation indicator, an outcome indicator, their interaction, Player B's beliefs about Player A's performance, and a vector of socio-demographic controls is in preparation and will appear in Appendix Table 5 once the analysis pipeline is in place; the unconditional comparison above implies that the coefficient on delegation, controlling for outcome, will be small and statistically insignificant.

The fact that we cannot reject equal punishment for delegated and self-made decisions is consistent with our specific design: because the algorithm is calibrated at the individual level and

Figure 4: Punishment by Player B and beliefs about punishment by Player A



Note: Average punishment chosen by Player B (solid bars) and average belief about punishment held by Player A (hollow bars) for each of the four scenarios. Bars are reported in pounds and standard errors are reported as error bars. Beliefs are elicited prior to learning the realised outcome and on the same scale as the punishment decision.

this calibration is common knowledge, the delegation decision in our setting carries no signal about Player A's belief in her own ability relative to the algorithm. A natural intention-based punishment motive is therefore muted: Player B has no obvious basis on which to read the delegation decision as overconfident, lazy, or selfish. By the same token, however, this design choice limits the test of H2 to a setting in which the most studied path through which delegation might insulate from punishment—signal extraction by the third party—is closed off by construction. The complementary reading is that responsibility shifting through delegation may operate more strongly when relative performance is uncertain (as in Feier et al. 2022); we discuss in Section 5 why a follow-up study that varies relative-performance calibration directly is the natural way to map this margin.

Beliefs and consistency

After making the delegation decision but before learning the outcome, we elicited Player A's beliefs about Player B's punishment in each of the four scenarios. Although the belief measure is unincentivised and noisy, two patterns emerge cleanly. First, Player As substantially *overestimate* the level of punishment they will face: average expected punishment exceeds average realised punishment in every scenario (Figure 4, hollow bars). Second, however, Player As correctly anticipate the *relative* pattern of punishment: they expect no significant difference in punishment between delegated and non-delegated decisions, conditional on outcome—a pattern that closely tracks the actual punishment behaviour of Player B.

This second pattern is consequential for the interpretation of Result 1: if Player A had reduced delegation in the *Punishment* because she anticipated being punished more harshly for delegation than for self-made decisions, we would expect to see this asymmetry in the elicited beliefs. We do not. The mechanism behind Result 1 is therefore unlikely to be a simple anticipation of differential punishment; we develop this argument in Section 4.

Within-subject consistency. Within the *Punishment* condition, the within-subject relationship between belief differences and the delegation decision goes in the predicted direction. Column (3) of Table 1 reports a Logit regression of delegation on Player A's belief difference (*no delegation* minus *delegation*) for each outcome, restricted to the *Punishment* condition. Coefficients on both belief differences are positive: a Player A who expects to be punished more for non-delegation than for delegation in a given outcome is more likely to delegate. The coefficient on the good-outcome belief difference is significant at conventional levels (1.58, two-sided $p = 0.033$); the coefficient on the bad-outcome belief difference is positive but noisier (0.94, two-sided $p = 0.21$). Despite the weakness of the elicitation, beliefs thus carry informational value: even at the individual level, those Player As who are more likely to delegate are those who expect, ex ante, to be punished more for not delegating.

Caveat on causal direction. We cannot, from this exercise alone, distinguish whether beliefs cause delegation or whether delegation causes (motivated) beliefs. Beliefs were elicited after the delegation decision; subjects may have rationalised their choice by holding consistent post-hoc beliefs. The unincentivised nature of the elicitation makes this even harder to rule out. The within-subject correlation is nonetheless consistent with the standard interpretation that anticipated punishment shapes delegation, while leaving the causal direction unresolved.

Mechanism: candidate explanations for the H1 reversal

The direction of Result 1 reverses the standard prediction of the responsibility-avoidance literature, in which the prospect of punishment should make delegation *more* attractive, not less. Because the reversal was not preregistered, the analyses in this subsection are exploratory and we treat the supported mechanisms as candidate hypotheses rather than as established. We consider four candidates and weigh each against the data, taking care to distinguish what we can rule out from what we can only flag as consistent.

(M1) Anticipated punishment for delegation. A Player A who expects to be punished more harshly for a delegated decision than for a self-made decision will rationally delegate less when punishment is on the table. *The data do not support this.* As shown in Section 4, Player As do not on average anticipate harsher punishment for delegated decisions; nor does Player B in fact deliver such punishment. A pure punishment-anticipation mechanism is therefore inconsistent with both the elicited beliefs and the realised punishment, and we view this candidate as ruled out by the available evidence.

(M2) Effort exertion to outperform the algorithm. An alternative mechanism is that the prospect of punishment motivates Player A to exert greater effort on the eleventh prediction, in the hope of personally producing the high payoff. Although our design controls for relative performance based on the first ten rounds—making the algorithm and Player A equivalent in expectation on the eleventh prediction—potential punishment could in principle shift Player A's effort margin. *The data are inconsistent with the simple form of this story.* Among Player As who chose not to delegate, those in the *No-Punishment* condition (where punishment is not possible) outperformed those in the *Punishment* condition: 45.5% of non-delegating Player As in the *No-Punishment* condition earned the high payoff for Player B, compared to 21.7% in the *Punishment* condition. This difference is not driven by ex-ante performance differences across the first ten rounds (*No-Punishment*: 3.12/10; *Punishment*: 2.98/10; two-sided $p = 0.66$).[9] Counter-intuitively, increased effort thus appears concentrated among non-delegators in the *No-Punishment* condition, where punishment is *not* on the table. This pattern is incompatible with a story in which Player A in the *Punishment* condition tries especially hard to outperform the algorithm in order to avoid punishment.

(M3) Process ownership of the outcome. A third mechanism, suggested by the heterogeneity of the delegation decision with respect to Player A's performance in the *Punishment* condition (Section 4), is that Player As who have higher subjective probability of producing the high payoff for Player B prefer to bring about that good outcome themselves rather than via the algorithm. The intuition draws on the psychological-ownership literature: people derive utility from being the proximate cause of a good outcome for another, and this utility is asymmetric with respect to outcomes (Steffel et al., 2016). *The data are consistent with this mechanism, but the supporting evidence is thin.* The negative relationship between performance and delegation in the *Punishment* condition is what process-ownership predicts: better-performing Player As, who are more likely to deliver the high payoff, prefer to do so personally. The absence of this relationship in the *No-Punishment* condition is consistent with process ownership being amplified by the moral salience that the punishment possibility introduces. We are explicit, however, that the supporting evidence consists of a single heterogeneity pattern in a 60-subject sub-sample (Column (3) of Table 1) and a comparison of selected non-delegator success rates;

[9] Measuring performance change as the difference between the average absolute prediction error across the first ten rounds and the absolute prediction error on the eleventh prediction, we find that Player As in the *No-Punishment* condition improve their performance from rounds 1–10 to round 11, whereas Player As in the *Punishment* condition do not improve. Because all Player As across both conditions faced the same image on the eleventh prediction, this differential improvement cannot be attributed to task-specific variation. We caveat that the comparison is between selected non-delegator subsamples, not between random subsets of all Player As; selection on the very outcome we are measuring is a known limit of this comparison.

the effect on either reading is the most consistent reading of the data we have, but not a confirmed mechanism. A direct test would require a treatment that varies process ownership while holding the punishment possibility fixed—for example, an arm in which the algorithm is presented as Player A's own "tool" rather than as a separate agent.

(M4) Experimenter demand. A final possibility is that the punishment possibility cues Player A to behave more conservatively—for instance, to avoid an action (delegation) that the experimental setting itself implicitly frames as morally questionable. *We cannot rule this mechanism out with our data.* The most we can say is that two implications of a strong demand story are not borne out: if demand were dominant, Player A's behaviour should track elicited beliefs more tightly than it does, and the sharp performance heterogeneity in the *Punishment* condition (better performers delegating less) would be hard to explain. Neither absence is decisive against the demand reading. We discuss the implications, and the design moves that would identify demand directly, in Section 5.

In sum: M1 (anticipated differential punishment) is inconsistent with the data. M2 (effort to outperform the algorithm) is inconsistent with the data in its simple form. M3 (process ownership) is consistent with the data but supported only by exploratory heterogeneity analysis. M4 (experimenter demand) cannot be separated from M3 with the present design. The most economical reading—the one we adopt with explicit hedges in the Conclusion—is that the morally charged context introduced by the punishment possibility makes Player A want to be the proximate cause of any good outcome for Player B, while we acknowledge that an experimenter-demand story can produce qualitatively similar patterns. Disentangling the two is the cleanest direction for follow-up work.

5 Discussion and Conclusion

This paper asks two questions about how the prospect of punishment interacts with the use of algorithms as decision aids. Can decision-makers shift responsibility for bad outcomes onto algorithms by delegating their decisions? And does the anticipation of being held responsible motivate delegation in the first place? We address both questions in a preregistered online experiment in which Player A makes a payoff-relevant prediction for Player B and may delegate the prediction to an algorithm calibrated to perform as well as Player A herself. A between-subject manipulation switches off punishment by Player B in the *No-Punishment* condition; the *Punishment* preserves the natural setting in which the prospect of punishment is on the table.

Two findings emerge. First, and most striking, the *prospect* of punishment *reduces*, rather than increases, the propensity to delegate to the algorithm: 42.5% of Player As delegate when punishment is possible, compared to 59.3% when it is not. This direction is the opposite of what the responsibility-avoidance literature would predict, and we view the reversal as the paper's main empirical contribution—noting that, because the result reverses the preregistered direction, the analyses we use to interpret it (Section 4) are exploratory rather than confirmatory. Second, conditional on the outcome for Player B, we cannot reject the hypothesis that delegated and self-made decisions are punished equally. The point estimate runs in the predicted direction (lower punishment under delegation in the bad-outcome cell) but is small relative to the test's confidence interval at the achieved sample size; we therefore frame Result 2 as a failure to detect insulation rather than as positive evidence that delegation cannot insulate. Player A does not appear to be able to use this particular algorithm as a meaningful blame shield in the bounded sense the test can assess.

Mechanism. Why does punishment reduce, rather than increase, delegation? The data are inconsistent with a simple anticipation mechanism: Player As do not on average expect harsher punishment for delegated decisions, and they correctly anticipate the absence of differential punishment that we observe in Player B's actual choices. The data are likewise inconsistent with the simple form of an effort story in which the prospect of punishment motivates Player A to outperform the algorithm: among non-delegators, those in the *No-Punishment* condition (where punishment is impossible) outperform those in the *Punishment* condition, the reverse of what such a story implies. The reading most consistent with the data is one of *process ownership*: Player As who actively choose not to delegate take responsibility for the outcome, exert correspondingly higher effort, and that share is higher when punishment is on the table. The supporting evidence is exploratory rather than confirmatory, however, and rests on a single performance-by-delegation heterogeneity pattern in a 60-subject sub-sample (Column (3) of Table 1); we cannot, with this design, separate process ownership cleanly from an experimenter-demand reading in which the punishment cue itself nudges Player A toward direct engagement. We therefore present process ownership as the most consistent reading of the data we have, not as a confirmed mechanism. Disentangling it from demand is a clean direction for follow-up work, sketched below.

This refinement is consistent with, but goes beyond, the standard reading of the responsibility-avoidance literature. The classic finding—that delegation reduces punishment of the principal and that this prospect motivates delegation—was established in dictator-game settings where the underlying choice is between a known fair and a known unfair allocation, the intermediary is human, and the principal can be held accountable specifically for choosing a particular intermediary (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013). When the intermediary is an algorithm calibrated to mimic the principal and the underlying decision involves genuine uncertainty, none of the standard mechanisms through which delegation can shift responsibility—signal value of the delegation choice, the human-human punishment channel, the certainty of the unfair outcome—are available. What remains is a baseline outcome-based punishment, supplemented by an asymmetric desire to be the proximate cause of good outcomes when the moral stakes are salient.

Implications, with explicit bounds. Our findings speak to the design of accountability regimes governing algorithmic decision-making. The European Union's AI Act (European Union, 2024) and analogous regulatory frameworks elsewhere place increasing emphasis on assigning responsibility to human users of high-risk algorithmic systems. The implicit theory of behaviour underlying these provisions is that holding humans accountable for algorithm-mediated decisions *disciplines* their use of those algorithms. Our results offer a behavioural complement to this view: when punishment is on the table, decision-makers in our setting do not simply delegate-and-blame the algorithm; they choose to take direct responsibility for the decision more often. From the perspective of efficiency, this need not be welfare-increasing: in domains where algorithms genuinely outperform humans, holding humans liable for outcomes may push decision-makers away from the more accurate technology. The trade-off between encouraging algorithmic adoption and assigning meaningful responsibility is, on this reading, sharper than the existing literature has suggested.

We note explicitly the bounds of this reading. Our evidence is a single online experiment with 322 UK Prolific participants on a stylised weight-prediction task, with stakes of £5 for Player B and a punishment range of £0–£2. The high-stakes algorithmic domains that motivate the regulatory debate—medical diagnosis, judicial risk assessment, hiring, credit allocation—differ from our setting in stake magnitude, professional context, expert training, the institutional structure of accountability, and the kind of population at the decision margin. We therefore view the result as identifying a behavioural channel that exists in principle and is detectable in our setting, not as a calibration of how big a lever accountability is in any specific applied domain. Mapping the magnitude of the effect against external benchmarks—e.g., physician adoption of diagnostic algorithms under malpractice exposure, or loan-officer deviation from credit-scoring algorithms under fair-lending audit—would require either domain-specific replication or instruments to which we do not have access. We flag this gap explicitly so that readers can calibrate the policy implication appropriately.

Limitations. Several caveats are in order. The belief measure we elicit is unincentivised and noisy; it conveys ordinal patterns reliably but cannot support fine-grained quantitative claims about Player A's anticipations. Two of the four belief scenarios are hypothetical conditional on the delegation decision, and all four are hypothetical in the *No-Punishment* condition; this further limits the inference, and we plan to report the H2 patterns separately for the payoff-relevant and the hypothetical cells once the analysis pipeline is in place. We use the strategy method for Player B's punishment decision, which is standard in this literature but may yield different absolute levels of punishment than a direct elicitation. Player A also makes the eleventh prediction whether or not she ultimately decides to delegate, which equates the time cost of the two options but may signal her commitment differently than a design in which the prediction is skipped under delegation. We did not include a separate manipulation-comprehension check verifying that Player A internalised the punishment-on/off distinction; we relied on the prominent screen-level framing and on attention-check pass rates as indirect evidence, but a future replication should add this check. The absolute stakes are modest (£5 high payoff, £0–2 punishment range): we cannot speak to how the effect scales with stakes, and a stake-magnitude robustness check is the first item in the future-research list below. Our sample is drawn from UK Prolific participants, which limits external validity along the dimensions in which Prolific samples differ from broader populations and from professional decision-makers in the high-stakes algorithmic domains we discuss above.

A natural concern is whether the punishment possibility cues Player A to behave more conservatively for reasons unrelated to the substantive question—an experimenter-demand effect. We cannot rule this concern out from the data. The direction of any such effect is also ambiguous: a decision-maker who is reminded of being judged could plausibly delegate *more* (to shed responsibility) or *less* (to demonstrate effort and commitment); the literature does not unambiguously favour one over the other. We note two patterns weakly inconsistent with a strong demand reading—behaviour does not track elicited beliefs as tightly as a pure-demand story would predict, and the sharp performance heterogeneity in the *Punishment* condition (better performers delegating less) is harder to motivate from demand alone—but we treat these as suggestive rather than dispositive. A direct test in the spirit of de Quidt et al. (2018) would be the natural next step.

Directions for future research. Several extensions would strengthen the inference. A direct-method companion would test the robustness of Result 2 against the strategy-method elicitation, and a meaningfully larger sample would let us put a tight confidence interval on the small but persistent point estimate of insulation under delegation. A condition in which Player A is told explicitly that Player B is unaware of the equal-performance calibration would test whether the relative-performance signal is the binding margin distinguishing our results from Feier et al. (2022). Conditions that experimentally vary the algorithm's performance relative to Player A (above, equal, below) would map the generality of the process-ownership reading. A demand-effect probe along the lines of de Quidt et al. (2018) would put a credible upper bound on the contribution of M4. A stake-magnitude robustness check, varying the £5/£1 high/low payoffs and the £0–£2 punishment range, would speak to whether the effect scales with stakes in the direction that policy-relevant settings would imply. Finally, replication in a different country, in a different decision domain (e.g., a medical-triage or hiring vignette), or with a sample of professional decision-makers rather than online crowd-workers would speak directly to the external-validity bounds we describe above.

Taken together, our results suggest a more textured story than the one that has emerged from the existing literature on responsibility avoidance and algorithm aversion. Within the bounds of the setting we study, the capacity to use algorithms as a tool for shifting responsibility appears limited, and the prospect of being held responsible—rather than encouraging delegation as a hedge—appears to push decision-makers towards direct engagement with the consequential choice. Whether the same channel operates in the high-stakes algorithmic domains that animate the regulatory debate is an empirical question that this study can frame but cannot, on its own, answer.

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Appendix

A Additional Tables

A.1 Condition balance — Player A

Table 2: Condition balance — Player A

Variable	*Punishment*	*No-Punishment*	p-value
Age	39.026	37.877	0.552
Female	0.487	0.494	1.000
Socio-economic status	5.179	5.321	0.564
Went to uni	0.575	0.556	0.928
Technology score	2.087	2.284	0.322
Leadership position	0.362	0.333	0.824

Note: Mean values by condition for Player A, with p-values from two-sided t-tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale; *Went to uni* indicates a university degree; *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items; *Leadership position* indicates self-reported management or supervisory experience.

A.2 Condition balance — Player B

Table 3: Condition balance — Player B

Variable	*Punishment*	*No-Punishment*	p-value
Age	37.087	42.356	0.011
Female	0.588	0.493	0.314
Socio-economic status	5.425	5.110	0.204
Went to uni	0.688	0.481	0.013
Technology score	1.363	1.333	0.869
Leadership position	0.287	0.321	0.771

Note: Mean values by condition for Player B, with p-values from two-sided t-tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale; *Went to uni* indicates a university degree; *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items; *Leadership position* indicates self-reported management or supervisory experience.

A.3 Full regression results — delegation decision

A.4 Determinants of Player B's chosen punishment

B Preregistration and deviations

The experiment was preregistered prior to data collection. The preregistration document specifies the two hypotheses (H1 and H2 in the body), the sample-size target (80 subjects per

Table 4: Determinants of the delegation decision — full controls

	Dependent variable: *Delegation*		
	Full sample		*Punishment only*
	(1)	(2)	(3)
No-Punishment indicator	0.677	0.757	
	(0.320)	(0.334)	
Punishment beliefs: bad outcome			0.937
			(0.751)
Punishment beliefs: good outcome			1.579
			(0.741)
Task performance		−0.192	−0.609
		(0.128)	(0.236)
Age		−0.004	0.064
		(0.015)	(0.030)
Female		0.350	1.334
		(0.369)	(0.793)
Socio-economic status		−0.058	−0.003
		(0.116)	(0.219)
Went to uni		0.091	−0.433
		(0.364)	(0.707)
Technology score		−0.078	0.059
		(0.153)	(0.283)
Leadership position		0.676	−0.387
		(0.367)	(0.729)
Constant	−0.302	0.402	−1.842
	(0.226)	(0.930)	(2.103)
N	161	159	60
Pseudo R^2	0.020	0.049	0.246

Note: Logit estimates with robust (HC1) standard errors in parentheses. Two subjects in Column (2) are dropped due to missing socio-demographic data. Column (3) restricts to the *Punishment* condition and excludes subjects who failed the attention check on the belief-elicitation screen, as preregistered. The belief variables in Column (3) are within-subject differences in expected punishment between non-delegated and delegated decisions for each outcome. Following AEA editorial policy, significance stars are not reported in the table.

Table 5: Determinants of Player B's chosen punishment

	Dependent variable: *Punishment*
	Punishment (£)
	(1)
Delegated	0.001
	(0.054)
Bad outcome	0.260
	(0.079)
Delegated × Bad outcome	−0.078
	(0.052)
Age	−0.001
	(0.006)
Female	−0.065
	(0.143)
Socio-economic status	0.072
	(0.049)
Went to uni	−0.099
	(0.136)
Technology score	0.030
	(0.058)
Leadership position	0.060
	(0.163)
Constant	0.027
	(0.344)
N (subject-by-cell)	268
Adj. R^2	0.034

Note: OLS estimates of Player B's chosen punishment on a delegation indicator, a bad-outcome indicator, their interaction, and socio-demographic controls. Standard errors clustered at the Player B level (in parentheses). Sample: Player B subjects in the *Punishment* condition who passed the punishment-elicitation attention check, expanded over the four (delegation, outcome) cells of the strategy method. Two-sided p-values: *Delegated* $p = 0.989$; *Bad outcome* $p = 0.001$; *Delegated* × *Bad outcome* $p = 0.137$. Following AEA editorial policy, significance stars are not reported.

role per condition, motivated by an ex-ante power calculation reproduced in the registry), the randomisation procedure, the attention-check exclusion rule, and the test direction for each preregistered comparison.

For full transparency, the table below summarises every analysis reported in the body of the paper, the corresponding preregistered specification, and any deviation. We treat the H1 reversal in Section 4 as a confirmatory test of the preregistered direction (which it rejects), and all subsequent mechanism analyses (Section 4) as exploratory.

Table 6: Preregistered tests, reported tests, and deviations

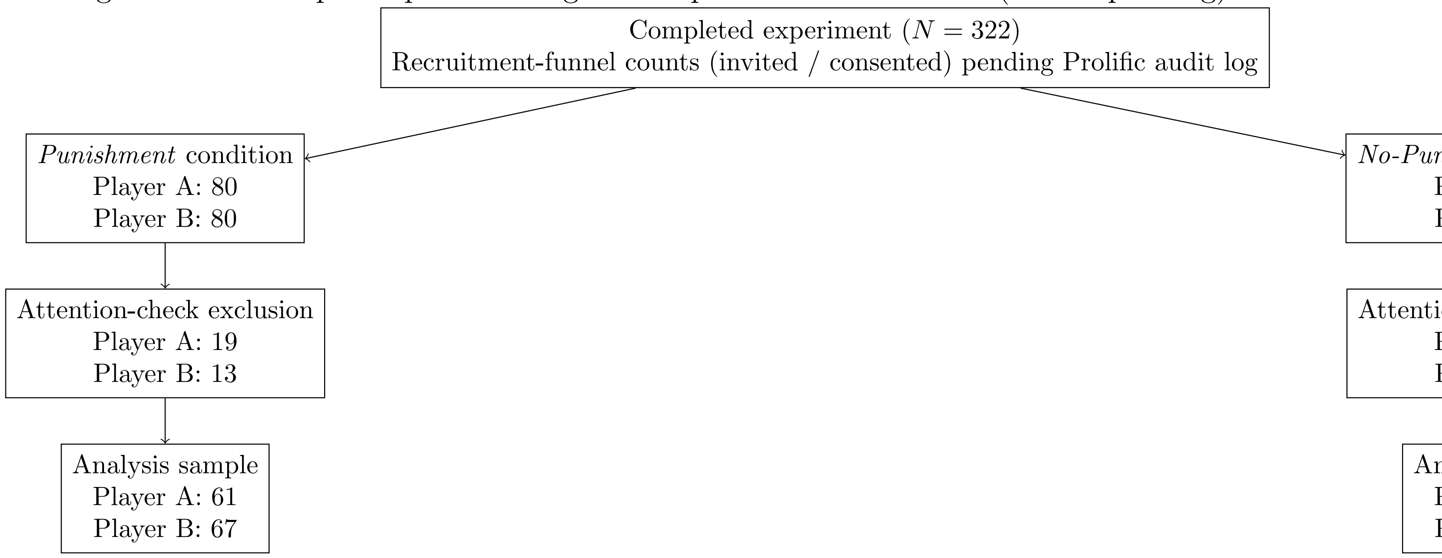
Preregistered specification	As reported	Deviation / rationale
H1: between-subject test of *Punishment* vs *No-Punishment* delegation rates; one-sided in the H1 direction (more delegation under *Punishment*)	Two-sided Chi-squared and Fisher's exact, $p = 0.049$ / 0.041	Direction reversed; one-sided test in H1 direction is non-significant by construction. We report two-sided to be transparent about what the data show.
H2: paired test of Player B punishment in the bad-outcome cell, delegated vs non-delegated, attention-check passers only	Paired t-test, $p = 0.153$; Wilcoxon signed-rank, $p = 0.533$	As preregistered.
Logit of delegation on condition + controls (Table 1, Cols (1)–(2))	As reported	As preregistered.
Logit on *Punishment* subsample with belief differences (Col (3))	As reported	Specification preregistered; sample restriction follows the preregistered attention-check rule.
Mechanism analyses M1–M4 (Section 4)	As reported	*Not preregistered.* Treated as exploratory throughout.
Effort-comparison statistics among non-delegators (Section 4, M2)	As reported	*Not preregistered.* Treated as exploratory.

Note: The preregistration document is available at *[registry link to be added at submission]*. The registry ID is *[ID to be added at submission]*.

C CONSORT-style flow diagram

A CONSORT-style flow diagram showing recruitment, attrition, attention-check exclusion, and final analysis sample size by condition and role is in preparation and will appear here once the analysis pipeline produces the underlying counts. The structure of the diagram is sketched below; cell counts pending.

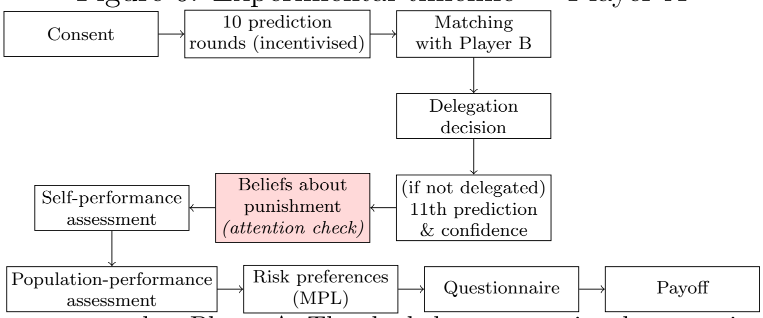
Figure 5: Flow of participants through the experiment — structure (counts pending)



D Additional Figures

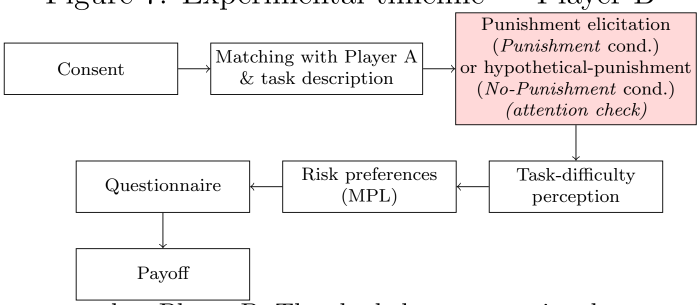
D.1 Experimental timelines

Figure 6: Experimental timeline — Player A



Note: Screen sequence presented to Player A. The shaded screen carries the preregistered attention check used to identify subjects excluded from belief-related analyses. The 11th-prediction-and-confidence screen is skipped if Player A delegates to the algorithm.

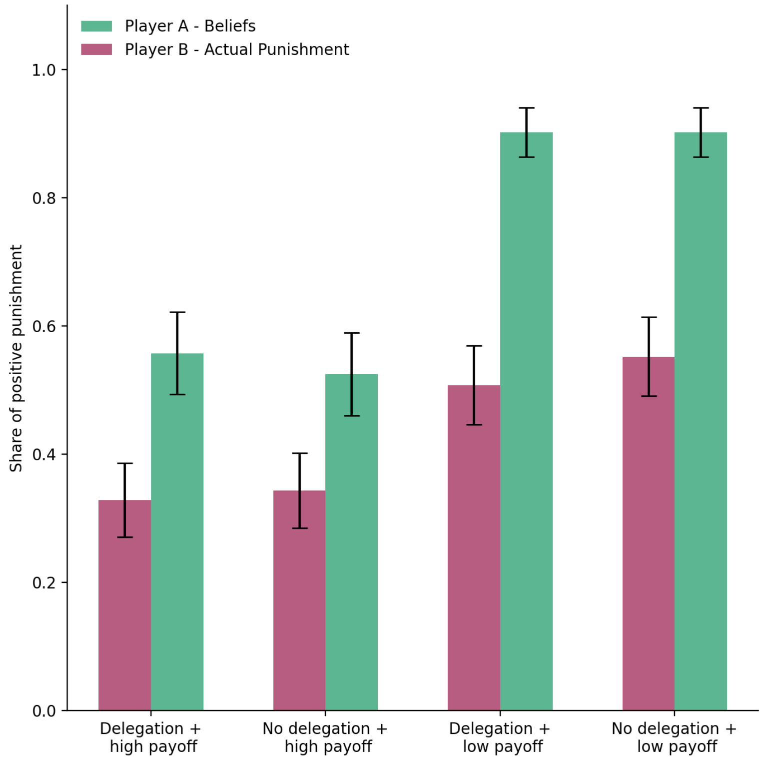
Figure 7: Experimental timeline — Player B



Note: Screen sequence presented to Player B. The shaded screen carries the preregistered attention check used to identify subjects excluded from punishment-related analyses. In the *No-Punishment* condition, no actual punishment is implemented; the elicitation is hypothetical and serves to maintain comparability of the screen sequence across conditions.

D.2 Punishment frequencies

Figure 8: Frequencies of punishment and beliefs about punishment



Note: Share of Player Bs choosing strictly positive punishment in each of the four scenarios (solid bars) and share of Player As assigning a strictly positive expected punishment in each scenario (hollow bars).